Course Schedule

2026-09-03

Course

Data Science Applications in Agriculture

"Beast & Bytes"

ANSC 4040/ ANSC 6040

Instructor

Dr. Miel Hostens (he/his) is the Robert and Anne Everett Endowed Associate Professor of Digital Dairy Management and Data Analytics at College of Agriculture and Life Sciences (Cornell University) focusing on the creation of methodologies using precision dairy farming to monitor sustainable food production systems from a global perspective

Office hours	Location
Quick questions	Slack & Teams

By appointment Thursday - Friday 9:00 am - 17:00 pm	Department of Animal Science, 273 Morrison Hall
Walk in office hours with Meike Wednesday 3 - 5 pm	Department of Animal Science, B30 Morrison Hall Please email me when this time does not work for you, so we can make an alternative arrangement.

Make an appointment with Dr. Hostens through this link.

Teaching assistants

	Name	Location
	Meike van Leerdam	B29 Morrison Hall
	Sonam Yang	B29 Morrison Hall

Course description

Description and aims

This course offers an an undergraduate (sophomore, junior, senior) and graduate level experience in challenge-based learning, where students tackle data-driven problems within the agricultural sector. If you have never heard about it, check out this Solar Car Challenge that you might recognize. Working in collaborative teams, students will engage directly with farmers and agricultural stakeholders to understand real-world challenges and develop innovative, data-driven solutions.

Throughout the course, students will learn to utilize a range of digital tools essential for modern data science. Key topics include data governance, data privacy, data collection, data storage, data processing, data visualization, data modeling, machine learning, artificial intelligence, and communicating data-driven results.

Project management is a core component of the course. Students will learn to manage projects effectively, adhere to deadlines, communicate with stakeholders, and deliver results in a timely manner. The course emphasizes teamwork, communication, collaboration, critical thinking, and responsible use of digital technologies.

By the end of the course, students will have gained practical experience in applying data science to agricultural problems. Students will develop skills in problem-solving, programming, data analysis, machine learning, artificial intelligence, project management, and communication.

The course also emphasizes the responsible and critical use of artificial intelligence. Students will learn to use AI tools efficiently while recognizing their capabilities, limitations, ethical implications, and potential effects on data privacy, academic integrity, and decision-making.

Credits

This course accounts for 3 credits.

Prerequisites

A statistical course such as one of the following is not required but strongly advised:

- STSCI 2150/5150 Introductory Statistics for Biology
- BTRY 3010/STSCI 2200 & BTRY 5010/STSCI 5200: Biological Statistics I
- ILRST/STSCI 2100: Introductory Statistics

Prior knowledge of Python is helpful but not required (you will learn vibe coding).

If in doubt about prerequisites, contact miel.hostens@cornell.edu

Learning objectives

1. **Understand the data science landscape** and its applications in agriculture.
2. **Develop practical coding skills** using Python and modern data science tools.
3. **Apply data science workflows and FAIR principles** to organize, manage, and analyze agricultural data.
4. **Scope agricultural problems using systems thinking** and translate them into data science projects.
5. **Independently design and execute a data science project**, from problem definition to results and communication.
6. **Reflect on their learning and take ownership of their development** as data science practitioners.
7. **Collaborate effectively in teams** to solve real-world agricultural problems.
8. **Use AI responsibly and efficiently**, recognizing its capabilities, limitations, and implications

Course Projects

Mini individual project

During the first half of the course, each student will complete a mini individual data science project.

The purpose of the mini project is to provide students with an opportunity to independently apply the tools and concepts introduced in the course.

Students will:

1. Identify an individual agricultural problem or question.
2. Define and scope the problem.
3. Identify and work with appropriate data.
4. Apply relevant data science methods.

5. Use appropriate computational and visualization tools.
6. Interpret and communicate their findings.
7. Document their workflow and decisions.
8. Present the results as a poster.

The specific project requirements and assessment criteria will be discussed during the course.

Final group project

In the second half of the course, students will engage in a group project involving a real-world, data-intensive problem in agriculture.

Working in teams, students will collaborate with farmers and stakeholders to identify and understand the problem. The team will then develop a data-driven solution using the tools and techniques learned during the course.

Project phases

1. Problem Identification: Meet with the farmer and other stakeholders to understand the problem and their needs. Conduct initial research to define and scope the problem.
2. Data Collection: Gather relevant data from appropriate sources while following data privacy, governance, and security requirements.
3. Data Analysis and Visualization: Analyze collected data and identify relevant patterns, trends, and insights.
4. Solution Development: Explore appropriate statistical, machine learning, or other data science approaches and develop a practical solution.
5. Implementation Plan: Develop a plan for implementing the proposed solution in an appropriate agricultural setting.
6. Presentation and Report: Present the methodology, findings, limitations, and proposed solution and submit the required project documentation.

Assessment and Grading

Assessment consists of individual assignments, a mini individual project with a poster presentation, and a final group project.

Individual assignments

Throughout the course, students will complete individual coding and practical assignments.

These assignments are designed to help students develop the technical skills needed for the mini individual project and final group project.

Individual assignments account for 20% of the final grade.

Assignment-specific instructions will identify any restrictions on collaboration, external resources, software, or AI tools.

Mini individual project and poster presentation

Each student will complete a mini individual project and present the results through a poster presentation.

The project provides an opportunity to apply the skills learned during the course to an individual topic or problem.

The mini individual project and poster presentation together account for 30% of the final grade.

The project deliverables, requirements, and assessment criteria will be discussed in class.

Final group project

The final assessment consists of a group project.

Students will work together to develop a larger project applying programming, data analysis, Git/GitHub, cloud-based tools, machine learning, and other data science skills learned throughout the course.

Together with the instructor and stakeholders, the group will define the goals and scope of the project.

The project should demonstrate:

- Effective collaboration
- Appropriate technical implementation
- Responsible data management
- Appropriate use of data science and AI tools
- Clear documentation

- Critical interpretation of results
- Clear communication of findings and limitations

The final group project accounts for 50% of the final grade.

The deliverables, project requirements, presentation format, and grading criteria will be discussed during the course.

Final grading

The final grade will be calculated as follows:

- 20% — Individual assignments
- 30% — Mini individual project + poster presentation
- 50% — Final group project

Total: 100%

Grade scale

Grade	Low	High
A+	99.80	100.0
A	93.33	99.80
A-	90.00	93.33
B+	86.66	90.00
B	83.33	86.66
B-	80.00	83.33
C+	76.66	80.00
C	73.33	76.66
C-	70.00	73.33
D+	66.66	70.00
D	63.33	66.66
D-	60.00	63.36
F	0.00	60.00

Each grade range includes the score on the left and excludes the score on the right. For example, a 90.0 is an A-, not a B+, while an 89.99 is a B+, not an A-.

Course Schedule

The detailed course schedule is maintained on the course website and may be updated during the semester.

The current planned schedule includes:

Week	Date	Activity
1	8/24	Course Outline
1	8/25	Toolbox: IDEs, Programming Languages & Vibe Coding
1	8/26	Data Science in Agriculture
2	8/31	Version Control Systems & Integration
2	9/1	Toolbox: Broken Online, Merge Conflicts & Branching
2	9/2	Naming Conventions & FAIR Principles
3	9/7	No Class – Labor Day Holiday
3	9/8	Cloud Computing, APIs & Hardware
3	9/9	Data Governance & Data Lineage
4	9/14	ML and AI Basics (Part 1)
4	9/15	Toolbox: Project Mini, Packages, Hugging Face & Model Serving
4	9/16	Ethical and Efficient Use of AI
5	9/21	ML and AI Basics; Supervised and Unsupervised Learning in Agricultural Contexts (Part 2)
5	9/22	Work on Mini Individual Project
5	9/23	Q&A / Introduction to Tableau
6	9/28	ML and AI Basics; Supervised and Unsupervised Learning in Agricultural Contexts (Part 3)

Week	Date	Activity
6	9/29	Work on Mini Individual Project
6	9/30	Q&A
7	10/5	Q&A
7	10/6	Poster Presentation
7	10/7	DISC Assessment of the Team
8	10/12	No Class – Fall Break
8	10/13	No Class – Fall Break
8	10/14	TBD
9	10/19	Final Project Kickoff
9	10/20	Farm Visit
9	10/21	Project Planning
10	10/26	Final Project
10	10/27	Final Project
10	10/28	Final Project
11	11/2	Final Project
11	11/3	Final Project
11	11/4	Final Project
12	11/9	Final Project
12	11/10	Final Project
12	11/11	Final Project
13	11/16	Final Project
13	11/17	Final Project
13	11/18	Final Project
14	11/23	Final Project
14	11/24	Final Project
14	11/25	No Class – Thanksgiving Break
15	11/30	Final Project Presentations / Submission Preparation
15	12/1	Final Project Submission
15	12/2	Course Wrap-up & Reflection

Policies

Attendance and Participation

Active participation is an important part of this challenge-based course.

Students are expected to attend lectures, labs, discussions, project meetings, and other scheduled course activities.

Because substantial portions of the course involve hands-on work, collaboration, and interaction with instructors and stakeholders, regular participation is important for successful completion of the course.

If circumstances prevent you from participating in a scheduled activity, communicate with the instructor or course staff as early as possible.

Inclusive community

I grew up in a family in which diversity, equity, and inclusion were important parts of everyday life. These values continue to shape how I approach teaching and learning.

I aim to ensure that students from diverse backgrounds and perspectives are well served by this course. I strive to address students' learning needs in and outside the classroom and to view the diversity that students bring as a resource, strength, and benefit.

My goal is to present materials and activities that respect diversity and align with Cornell University's core values.

Your suggestions are truly encouraged and appreciated. Please let me know how I can improve the course's effectiveness for you personally or for other students and student groups.

I also aim to foster a learning environment that embraces diversity of thought, perspectives, and experiences and respects the identities of all members of the course community.

If experiences outside of class are affecting your performance, please feel free to speak with the instructor. Alternatively, your academic dean is a resource if you prefer to speak with someone outside the course.

Mental Health and Stress management

If you are feeling overwhelmed, or are worrying about a friend, please reach out to me, one of your instructors or your academic adviser.

We can try to help, or we can put you in touch with someone who can help. Cornell has trained counselors available to listen and help: Cornell Health's Counseling and Psychological Services (CAPS, 607-255-5155); [Student Support and Advocacy Services] (https://scl.cornell.edu/student-support); Empathy, Assistance, and Referral Service (EARS, 213 Willard Straight Hall, 607-255-3277), and Let's Talk.

The Learning Strategies Center offers a range of academic resources.

Academic Integrity

Absolute integrity is expected of every Cornell student in all academic undertakings. Integrity entails a firm adherence to values grounded in honesty with respect to the intellectual efforts of oneself and others and in free and open inquiry and discussion.

Academic integrity applies to all academic work and interactions connected to the educational process, including the use of University resources and digital technologies.

A Cornell student's submission of work for academic credit indicates that the work is the student's own. All outside assistance should be appropriately acknowledged, and students are responsible for truthfully representing their academic work.

Students are expected to follow the Cornell University Code of Academic Integrity, including the version effective for Fall 2026.

Students should review the University's Guidelines for Students and ask the instructor or teaching assistants whenever they are uncertain whether a particular activity is permitted. See also: https://undergrad.cornell.edu/students/academic-integrity/essential-guide-academic-integrity-undergraduate-students

Academic integrity expectations apply regardless of whether work is completed using traditional resources, online resources, programming tools, generative AI, or other technologies.

Each person in this class is expected to respect the principles of academic freedom for instructors and classmates and will maintain the privacy of the classroom environment, as outlined in Cornell's S20 Commitment to Academic Integrity, Equitable Instruction, Trust, and Respect.

Responsible Use of Generative AI

Generative artificial intelligence and other generative tools are a useful part of this course and may be used as learning and development aids unless an assignment states otherwise.

Appropriate uses include learning concepts, brainstorming, programming assistance, debugging, exploring technical and data-analysis approaches, generating practice examples, improving student-written material, and learning to work with APIs, packages, models, and other technical tools.

Students remain responsible for the accuracy, originality, interpretation, security, and integrity of their submitted work and should understand and be able to explain anything they submit.

AI-generated code, analysis, calculations, citations, references, facts, claims, and technical explanations should be carefully reviewed and independently verified.

When AI materially contributes to submitted work, students should follow the assignment's requirements for attribution and documentation. AI use does not override Cornell's Code of

Academic Integrity, and students should ask the instructor or teaching assistants when they are unsure whether a particular use is appropriate.

See also: https://it.cornell.edu/ai-strategy/ai-guidelines , https://teaching.cornell.edu/generative-artificial-intelligence , https://teaching.cornell.edu/generative-artificial-intelligence/ai-academic-integrity

Privacy, Confidentiality, and Responsible Data Use

Students should be especially careful when using generative AI or other external tools with course and project data.

Confidential farmer information, personally identifiable information, proprietary datasets, credentials, passwords, private research data, and other sensitive information should not be uploaded to AI systems or external services unless the instructor has confirmed that the tool and use are appropriate.

Students are responsible for following applicable data-use agreements, project-specific data governance requirements, and appropriate security practices.

AI-assisted work should be tested, validated, documented, and critically evaluated, and teams remain responsible for their code, analysis, interpretation, sources, and conclusions. External and AI-generated sources, data, code, images, and other contributions should be appropriately acknowledged and verified.

Accommodations for Students with Disabilities

Students with disabilities who require accommodations should contact Student Disability Services (SDS) and provide the appropriate accommodation documentation as early as possible in the semester.

If you have an approved accommodation, please communicate with the instructor so that appropriate arrangements can be made.

Student Disability Services can be contacted through Cornell's SDS office.

Student Support

If you are experiencing difficulties that affect your ability to participate or succeed in the course, please communicate with the instructor or your academic adviser.

Cornell provides a range of academic, health, counseling, accessibility, and student-support resources. Students are encouraged to seek assistance early rather than waiting until difficulties become overwhelming.

Course Communication

The course websiteand Canvas will contain the most up-to-date information about course activities, assignments, project requirements, and schedule changes.

Course announcements and important changes may also be communicated by email.

Students are responsible for checking the course website and course communications regularly.

Final Note to Students

This course is designed to help you develop the skills needed to work with data, computational tools, artificial intelligence, and agricultural problems in a rapidly changing professional environment.

You are not expected to know everything at the beginning of the course. You are expected to learn, experiment, ask questions, collaborate appropriately, critically evaluate your tools and results, and take responsibility for your own learning.

When you are uncertain about a technical, academic-integrity, collaboration, data-privacy, or AI-use question, please feel free to reach out. The teaching team is here to help you make responsible decisions and develop the skills needed to work effectively and ethically as a data science practitioner.

As this is a continuously evolving, challenge-based course, some activities, policies, and project details may be updated during the semester.

We will communicate any significant changes through Canvas and keep the course website up to date with the latest course materials, schedule, and requirements.